

Lukas Brunner

curriculum vitae

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Current Position

05/2024 - **Senior researcher**
present Cluster of Excellence Climate, Climatic Change, and Society (CLICCS), University of Hamburg

Previous Positions

01/2022 - **Senior scientist**
04/2024 Group for Climate Dynamics and Modeling, Department of Meteorology and Geophysics, University of Vienna
05/2023 - **Visiting scientist**
06/2023 Institute of Meteorology, Freie Universität Berlin
10/2020 - **Senior scientist**
12/2021 Climate Physics group, Institute for Atmospheric and Climate Science, ETH Zurich
05/2018 - **Postdoctoral scientist**
09/2020 Climate Physics group, Institute for Atmospheric and Climate Science, ETH Zurich
04/2017 - **Visiting scientist**
09/2017 Center for International Climate Research Oslo (CICERO)
10/2015 - **Visiting scientist**
03/2016 Chair of Climate System Science, School of Geosciences, University of Edinburgh
10/2014 - **Predoctoral scientist**
03/2018 Research group ARSCliSys, Wegener Center, University of Graz

Education

04/2025 **Habilitation thesis submitted**, University of Hamburg, Germany
Thesis Evaluating climate model performance in multi-model ensembles
03/2018 **Dr. rer. nat. in Climate physics**, University of Graz, Austria
Thesis A new perspective on atmospheric blocking – detection, analysis, and impacts
07/2014 **MSc. in Geophysics**, University of Graz, Austria
Thesis Stratospheric ozone and temperature evolution over the past decades
10/2012 **BSc. in Physics**, University of Graz, Austria
06/2008 School leaving examination

Funded Projects

2024 **High-resolution machine learning for the climate community in Austria**
Austrian Climate Research Programme; 300.000 Euro, June 2024 to May 2027
<https://klimaszenarien.at/projekte/datennutzbarkeit/>

Committee Work, Convening, and Volunteering

- since 2026 **Member of the CLICCS Diversity, Equity, and Inclusion Board**
- since 2024 **Member Scientific Advisory Board Austrian Climate Scenarios**
- 2025 **Team lead at the WCRP global km-scale hackathon**
WCRP Digital Earths Lighthouse Activity global km-scale hackathon
- 2025 **Session convener at EGU25**
 - New developments in Earth system modelling: model evaluation, benchmarking, and constraining
 - Up-Goer Five Challenge: Making Big Ideas Simpler by Talking About Them in Words We Use a Lot
- 2024 **Session convener at EGU24**
Regional Climate Modelling, Including CORDEX and Constraining Global Multi-Model Ensembles
- 2024 **Mentor EGU mentoring programme**
- 2023 - **Member University of Vienna sustainability advisory board**
- 2024 University of Vienna
- 2022 - **Member of the steering committee of the Austrian Climate Scenarios**
- 2024 Klimaszenarien.AT
- 2021 - **Climate expert and participant at the Future of the Earth performance**
- 2022 The Future of the Earth interactive theatre performance
- 2022 **Session chair CliMatters workshop**
Uncertainty in climate modelling and projections
- 2019 - **Board member (president in 2021)**
- 2021 Club Alpbach Vorarlberg, European Forum Alpbach
- 2018 - **Reviewer for scientific journals**
- present 50+ verified reviews on WoS, including for J. Climate, Nature, and Science Advances

Awards and Scholarships

- 2023 **Wissenschaftspreis des Landes Vorarlberg for young researchers**
Vorarlberger Landesregierung
- 2022 **Scholarship as youth representative of the European Forum Alpbach at COP27**
European Forum Alpbach Foundation and Forum Alpbach Network
- 2018 **Performance scholarship for my PhD project**
University of Graz
- 2017 - **Member PRO SCIENTIA**
- 2018 Österreichisches Studienförderungswerk
- 2016 - **Scholarships for the European Forum Alpbach**
- 2018 Club Alpbach Vorarlberg: 2016 & 2017; European Forum Alpbach Foundation: 2018
- 2016 **Marietta Blau scholarship**
Austrian Exchange Service

Key Presentations

- 05/2025 EGU General Assembly, Vienna
- 04/2024 WCRP, APARC ATC spring meeting, Graz
- 04/2024 MPI-M/University of Hamburg, Klimacampus Kolloquium, Hamburg
- 03/2025 University of Bern, Colloquium in Atmosphere, Climate and Earth Observation, Bern
- 01/2025 University of Innsbruck, ACINN Graduate Seminar, Innsbruck
- 10/2024 NCKF Climate Research Symposium, Copenhagen (keynote)
- 07/2024 Universität Freiburg, Freiburg
- 04/2024 EGU General Assembly, Vienna
- 03/2024 Deutsche Klimatagung, Potsdam (keynote)
- 02/2024 Climate Analytics, Berlin
- 07/2023 IUGG General Assembly, Berlin
- 05/2023 Colloquium Leipzig Institute for Meteorology, Leipzig University, Leipzig
- 05/2023 Institute for Meteorology colloquium, FU Berlin, Berlin
- 05/2023 Climate statistics and extremes group, University of Hamburg, Hamburg
- 03/2023 Global Carbon Budget workshop, online (keynote)
- 05/2022 EGU General Assembly, Vienna
- 01/2022 Colloquium Department of Meteorology and Geophysics, University of Vienna, Vienna
- 11/2021 EC-Earth Consortium General Assembly 2021, online (keynote)
- 10/2021 Wegener Center Common Space, University of Graz, Graz
- 04/2017 EGU General Assembly, Vienna
- 12/2015 AGU Fall Meeting, San Francisco

Supervision and Teaching

Supervision

- ongoing **Using km-scale global climate model simulations to study the physical impacts of climate change for informing adaptation strategies** (working title)
Supervision of PhD thesis with Jana Sillmann
- ongoing **Effect of an AMOC shutdown on extremes in Europe using km-scale models** (working title)
Co-supervision of Master's thesis with Keno Riechers (MPI-M)
- 2025 **Spatiotemporal Roles of Heatwave Drivers in Europe**
Co-supervision of Master's thesis with Jana Sillmann
- 2025 **Separating Sources of Uncertainty in Future Climate Extremes**
Co-supervision of Master's thesis with Jana Sillmann
- 2024 **Effect of climate variability and extremes on heating and cooling energy demand in Canada**
Co-supervision of Master's thesis with Jana Sillmann
- 2024 **Exploring Classifier Skill in Distinguishing Climate Model and Observational Data**
Supervision of Master's thesis
- 2024 **Changes in heatwave properties under climate change**
Supervision of Master's thesis with Aiko Voigt

- 2024 **Extracting forced signals of climate change from daily temperature patterns across scales with CESM2**
Supervision of Bachelor's thesis
- 2024 **Analyse eines Fehlers in der Definition der Hitzemetrik Cumulative Heat**
Supervision of Bachelor's thesis
- 2021 **Interannual to decadal precipitation variability in a warming climate: spatial structure and underlying mechanisms**
Co-supervision of Master's thesis with Flavio Lehner
- 2020 **Assessing climate model uncertainties – an encounter with CMIP6**
Supervision of Bachelor's thesis
- 2019 **Changes in northern hemisphere blocking occurrence under 2×CO₂ based on CESM**
Supervision of Bachelor's thesis
- [Teaching](#)
- since 2025 **Basics of climate change**
Bachelor course, University of Hamburg (part of the course)
- since 2024 **Weather and Climate Extremes in a changing climate**
Master course, University of Hamburg (part of the course)
- 2024 **Climate Modelling**
Master course, University of Hamburg (part of the course)
- 2023 - **Paper club**
2024 Master course, University of Vienna (part of the course)
- 2022 - **Numerische Methoden**
2024 Bachelor course, University of Vienna (full course)
- 2022 - **Modelling and Data Analysis**
2024 Master course, University of Vienna (part of the course)
- 2022 - **Exercises in Introduction to Computational Meteorology**
2024 Master course, University of Vienna (full course)

Selected Outreach Activities and Media Mentions

- 2025 Public lecture: Wie ändern sich Wetterextreme mit dem Klimawandel?
<https://youtu.be/OKGi4Y6J2Ls>
- 2025 School lecture: Wetter- und Klimaextreme im Kontext des Klimawandels
- 2024 Guest lecture: Klimarecht im intra- und interdisziplinären Dialog, Faculty of Law, University of Vienna
- 2024 Guest lecture: Green Finance Seminar, Faculty of Law, University of Vienna
- 2024 Mentioned in Media: Die Presse
"Irrtum der Experten: Es gibt mehr Hitzetage", <https://www.diepresse.com/18290185/irrtum-der-experten-es-gibt-mehr-hitzetage>
- 2023 Open debate: Food for thought, European Forum Alpbach
"Data sparks action: the role of information in tackling the climate crisis"; together with David Jablonski
- 2023 Guest lecture: Green Finance Seminar, Faculty of Law, University of Vienna
"Naturwissenschaftliche Grundlagen der Klimakrise"

- 2022 Climate expert and participant at the Future of the Earth interactive theatre performance, Basel
- 2022 Public lecture: Inspirationstag der Katholischen Kirche Wien
“Naturwissenschaftliche Grundlage der Klimakrise”
- 2022 Interview in Ö1 public radio
“Wie entstehen Wolken?”, <https://oe1.orf.at/programm/20221027/695314/Wie-entstehen-Wolken>
- 2022 Participation at the Vienna Kinderuni
- 2022 Article: Rudolphina
<https://rudolphina.univie.ac.at/klimakonferenz-cop27-hitze-duerre-flut-extremwetter-katastrophen>
- 2021 Climate expert at the Future of the Earth interactive theatre performance, Basel
- 2020 Mentioned in media: Ars Technica
<https://arstechnica.com/science/2020/11/newest-climate-models-shouldnt-raise-future-warming-projections>
- 2017 Mentioned in media: Der Standard (in German)
<https://www.derstandard.at/story/2000051806269/atmosphaerische-blockaden-fuehren-zu-kaelteeinbruechen-im-fruehjahr>

Publications

- in review Uckan Y., **Brunner L.**, Sillmann J., and Kang S. M.: Influence of Southern Ocean temperature trend biases on fall cold extremes in North America, *Clim. Dyn.*, DOI: <https://doi.org/10.21203/rs.3.rs-9610006/v1>
- in review van Heerwaarden C. C. et al. (with **Brunner L.**): Evaluating local climate in global storm-resolving models with the Köppen-Geiger classification, *Environ. Res. Lett.*, DOI: <https://doi.org/10.48550/arXiv.2604.25447>
- in review Castillo Bautista N., Gaetani M., Borchert L. F., Poschlod B., **Brunner L.**, Sillmann J., and Martina M. L. V.: Large-scale drivers of compounding hot and dry events in three breadbasket regions, *EGUsphere*, DOI: <https://doi.org/10.5194/egusphere-2025-5073>
- 04/2026 **Brunner L.**, Ghosh R., Haimberger L., Hohenegger C., Putrasahan D., Rackow T., Knutti R., and Voigt A.: Three decades of simulating global temperature patterns with coupled global climate models, *Commun. Earth Environ.*, DOI: <https://doi.org/10.1038/s43247-026-03497-w>
- 02/2026 Holtanová E., Koláček J., and **Brunner L.**: A global perspective on past and future change in regional seasonal cycle shape, *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-17-181-2026>
- 02/2026 Yoon A., Hohenegger C., Bao J., and **Brunner L.**: Extreme events in the Amazon after deforestation, *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-17-167-2026>
- 10/2025 Segura H. et al. (with **Brunner L.**): nextGEMS: entering the era of kilometer-scale Earth system modeling, *Geosci. Model Dev.*, DOI: <https://doi.org/10.5194/gmd-18-7735-2025>
- 09/2025 Maher N., Phillips A. S., Deser C., Jnglin Wills R. C., Lehner F., Fasullo J., Caron J. M., **Brunner L.**, Beyerle U., and Jeffree J.: The updated Multi-Model Large Ensemble Archive and the Climate Variability Diagnostics Package: New tools for the study of climate variability and change, *Geosci. Model Dev.*, DOI: <https://doi.org/10.5194/gmd-18-6341-2025>
- 06/2025 **Brunner L.**, Poschlod B., Dutra E., Fischer M. E., Martius O., Sillmann J.: A global perspective on the spatial representation of climate extremes from km-scale models, *Environ. Res. Lett.*, DOI: <https://doi.org/10.1088/1748-9326/ade1ef>
- 05/2025 Sobolowski S., Somot S., Fernandez J., Evin G., Brands S., Maraun D., Kotlarski S., Jury M., Benestad R. E., Teichmann C., Christensen O. B., Bülow K., Buonomo E., Katragkou E., Steger C., Sørland S., Nikulin G., McSweeney C., Dobler A., Palmer T., Wilcke R., Boé J., **Brunner L.**, Ribes A., Qasmi S., Nabat P., Sevault F., Oudar T.: GCM Selection & Ensemble Design: Best Practices and Recommendations from the EURO-CORDEX Community, *BAMS*, DOI: <https://doi.org/10.1175/BAMS-D-23-0189.1>
- 11/2024 Nguyen V. D., Vorogushyn S., Nissen K., **Brunner L.**, and Merz B.: A non-stationary climate-informed weather generator for assessing future flood risks, *Advances in Statistical Climatology, Meteorology and Oceanography*, DOI: <https://doi.org/10.5194/ascmo-10-195-2024>
- 04/2024 O'Reilly C., **Brunner L.**, Qasmi S., Nogherotto R., Ballinger A., Booth B., Befort D., Knutti R., Schurer A., Ribes A., Weisheimer A., Coppola E., and McSweeney C.: Assessing observational constraints on future European climate in an out-of-sample framework, *npj Climate and Atmospheric Science*, DOI: <https://doi.org/10.1038/s41612-024-00648-8>

- 03/2024 **Brunner L.** and Voigt A.: Pitfalls in diagnosing temperature extremes, *Nature Communications*, DOI: <https://doi.org/10.1038/s41467-024-46349-x>
- 08/2023 Merrifield A. L., **Brunner L.**, Lorenz R., Humphrey V., and Knutti R.: Climate model Selection by Independence, Performance, and Spread (ClimSIPS v1.0.1) for regional applications, *Geosci. Model Dev.*, DOI: <https://doi.org/10.5194/gmd-16-4715-2023>
- 07/2023 **Brunner L.** and S. Sippel: Identifying climate models based on their daily output using machine learning, *Env. Data Sci.*, DOI: <https://doi.org/10.1017/eds.2023.23>
- 04/2023 Palmer T. E., C. F. McSweeney, B. B. Booth, M. D. K. Priestley, P. Davini, **L. Brunner**, L. Borchert, and M. B. Menary: Performance based sub-selection of CMIP6 models for impact assessments in Europe, *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-14-457-2023>
- 10/2022 Gründemann G. J, N. van de Giesen, **L. Brunner**, and R. van der Ent: Rarest rainfall events will see the greatest relative increase in magnitude under future climate change, *Commun. Earth Environ.*, DOI: <https://doi.org/10.1038/s43247-022-00558-8>
- 06/2022 Befort D. J., **Brunner L.**, Borchert L. F., O'Reilly C. H., Mignot J., Ballinger A. P., Hegerl G. C., Murphy J. M.: Combination of decadal predictions and climate projections in time: Challenges and potential solutions, *Geophys. Res. Lett.*, DOI: <https://doi.org/10.1029/2022GL098568>
- 06/2021 Hegerl, G., A. P. Ballinger, B. Booth, L. F. Borchert, **L. Brunner**, M. Donat, F. Doblas-Reyes, G. Harris, J. Lowe, R. Mahmood, J. Mignot, J. Murphy, D. Swingedouw, and A. Weisheimer: Toward Consistent Observational Constraints in Climate Predictions and Projections, *Front. Clim.*, DOI: <https://doi.org/10.3389/fclim.2021.678109>
- 11/2021 Sperna Weiland, F. C., R. D. Visser, P. Greve, B. Bisselink, **L. Brunner**, and A. H. Weerts: Estimating Regionalized Hydrological Impacts of Climate Change Over Europe by Performance-Based Weighting of CORDEX Projections. *Front. Water*, DOI: <https://doi.org/10.3389/frwa.2021.713537>
- 11/2020 **Brunner, L.**, A. G. Pendergrass, F. Lehner, A. L. Merrifield, R. Lorenz, and R. Knutti: Reduced global warming from CMIP6 projections when weighting models by performance and independence. *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-11-995-2020>
- 09/2020 **Brunner, L.**, C. McSweeney, A. P. Ballinger, D. J. Befort, M. Benassi, B. B. Booth, E. Coppola, H. de Vries, G. Harris, G. C. Hegerl, R. Knutti, G. Lenderink, J. Lowe, R. Nogherotto, C. O'Reilly, S. Qasmi, A. Ribes, P. Stocchi, and S. Undorf: Comparing Methods to Constrain Future European Climate Projections Using a Consistent Framework. *J. Climate*, DOI: <https://doi.org/10.1175/jcli-d-19-0953.1>
- 09/2020 Merrifield, A. L., **L. Brunner**, R. Lorenz, and R. Knutti: An investigation of weighting schemes suitable for incorporating large ensembles into multi-model ensembles. *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-11-807-2020>
- 05/2020 Lehner, F., C. Deser, N. Maher, J. Marotzke, E. Fischer, **L. Brunner**, R. Knutti, and E. Hawkins: Partitioning climate projection uncertainty with multiple Large Ensembles and CMIP5/6. *Earth Syst. Dynam.*, DOI: <https://doi.org/10.5194/esd-11-491-2020>
- 09/2019 **Brunner, L.**, R. Lorenz, M. Zumwald, and R. Knutti: Quantifying uncertainty in European climate projections using combined performance-independence weighting. *Env. Res. Lett.* DOI: <https://doi.org/10.1088/1748-9326/ab492f>

- 06/2018 Unterberger, C., **L. Brunner**, S. Nabernegg, K. Steininger, A. K. Steiner, E. Staben-theiner, S. Monschein and H. Truhetz: Spring frost risk for regional apple production under a warmer climate. *PLoS ONE*, DOI: <https://doi.org/10.1371/journal.pone.0200201>
- 01/2018 **Brunner, L.**, N. Schaller, J. Anstey, J. Sillmann and A. K. Steiner: Dependence of present and future European temperature extremes on the location of atmospheric blocking. *Geophys. Res. Lett.*, DOI: <https://doi.org/10.1029/2018GL077837>
- 01/2017 **Brunner, L.** and A. K. Steiner: A global perspective on atmospheric blocking using GPS radio occultation – one decade of observations. *Atmos. Meas. Tech.*, DOI: <https://doi.org/10.5194/amt-10-4727-2017>
- 11/2016 **Brunner, L.**, G. C. Hegerl and A. K. Steiner: Connecting atmospheric blocking to European temperature extremes in spring. *J. Climate*, DOI: <https://doi.org/10.1175/JCLI-D-16-0518.1>
- 04/2016 **Brunner, L.**, A. K. Steiner, B. Scherllin-Pirscher and M. W. Jury: Exploring atmospheric blocking with GPS radio occultation observations. *Atmos. Chem. Phys.*, DOI: <https://doi.org/10.5194/acp-16-4593-2016>

Other publications

- 03/2020 **Brunner, L.**, M. Hauser, R. Lorenz, and U. Beyerle: The ETH Zurich CMIP6 next generation archive: technical documentation. DOI: <http://doi.org/10.5281/zenodo.3734128>
- 06/2018 Mohankumar, S. E. P., K. Mintz-Woo, M. Damert, **L. Brunner** and J. Eise: Blogging Climate Change: A Case Study, In: Addressing the Challenges in Communication Climate Change Across Various Audiences. DOI: <https://doi.org/10.1007/978-3-319-98294-6>
- 04/2018 **Brunner, L.**: A new perspective on atmospheric blocking from observations – detection, analysis, and impacts (Dissertation). Wegener Center Verlag Graz, Scientific Report Nr. 76-2018, URL: <https://wegcwww.uni-graz.at/publ/wegcreports/2018/WCV-SciRep-No76-LBrunner-Jun2018.pdf>
- 06/2014 **Brunner, L.**: Stratospheric ozone and temperature evolution over the past decades (Master's thesis). Wegener Center Verlag Graz, Scientific Report Nr. 59-2014, DOI: http://wegcwww.uni-graz.at/publ/wegcreports/2014/WCV-SciRep-No59-LBrunner-Aug2014_1.pdf

Published Code and Data

- Dataset **Brunner L.** (2025): Annual ETCCDI Extreme Indices For ICON-Sapphire And IFS-FESOM (nextGEMS Cycle 4), https://doi.org/10.26050/WDCC/nextGEMS_ExtrInd
- Code **Brunner L.** (2025): Code for: A global perspective on the spatial representation of climate extremes from km-scale models, <https://doi.org/10.5281/zenodo.15619824>
- Code **Brunner L.** (2025): Code for: A global perspective on the spatial representation of climate extremes from km-scale models, https://github.com/lukasbrunner/Brunner_etal_2025_ERL
- Code **Brunner L.** (2024): Code for: Pitfalls in diagnosing temperature extremes, https://github.com/lukasbrunner/running_window_bias
- Dataset & Code **Brunner L.** and S. Sippel (2023): Data and code for Identifying climate models based on their daily output using machine learning. <https://doi.org/10.5281/zenodo.7998436>
- Dataset Liu, Y, P. Kalverla, F. Alidoost, S. Verhoeven, B. Vreede, B. Booth, E. Coppola, R. Nogherotto, **L. Brunner**, G. Harris, S. Qasmi, A. Ballinger, G. Hegerl, C. McSweeney, C. O'Reilly, T. Palmer, A. Ribes, and H. de Vries (2021): Pre-processed data of atlas in EUCP-WP2 (1.0.1). <https://doi.org/10.5281/zenodo.5679560>
- Code **Brunner L. et al**: Climate Model Weighting by Independence and Performance (ClimWIP)
<https://github.com/lukasbrunner/ClimWIP> (collaborative project)
https://docs.esmvaltool.org/en/latest/recipes/recipe_climwip.html
- Code **Brunner L.**: A global blocking detection algorithm
<https://github.com/lukasbrunner/blocking>
- Dataset **Brunner L.**, M. Hauser, R. Lorenz, and U. Beyerle (2020): The ETH Zurich CMIP6 next generation archive: technical documentation. DOI: <http://doi.org/10.5281/zenodo.3734128>
- Dataset **Brunner L.**: Gridded radio occultation satellite data
Geopotential Height: <https://hdl.handle.net/20.500.11756/e4f48220>, Temperature: <https://hdl.handle.net/20.500.11756/8245c63e>, Specific Humidity: <https://hdl.handle.net/20.500.11756/8245c63e>